Experiment 3: Incident Response Playbook for DDoS Attacks

Aim

The objective of this experiment is to develop a detailed incident response playbook to handle Distributed Denial of Service (DDoS) attacks effectively.

Inputs

This experiment considers a simulated DDoS attack scenario along with a basic network architecture setup.

Tools Used

A text editor is used for documenting the playbook, and diagram tools are used to represent the network architecture.

Dataset Required

The dataset includes sample indicators of DDoS attacks such as unusually high network traffic, repeated requests from multiple IP addresses, and sudden increases in bandwidth usage.

Brief Theory

A DDoS attack targets a system or network by flooding it with excessive traffic from multiple sources, causing service disruption.
An incident response playbook outlines predefined steps that guide security teams in responding to such attacks in a consistent and organized manner.

Procedure

Step 1: Detection Criteria

Network traffic is continuously monitored to identify abnormal traffic patterns. Alerts from security systems and signs of service degradation are used to detect a possible DDoS attack.

Step 2: Mitigation Steps

Appropriate mitigation actions such as traffic filtering, rate limiting, and blocking suspicious IP addresses are implemented to reduce the impact of the attack.

Step 3: Role Assignment

Specific responsibilities are assigned to team members to ensure smooth coordination during the incident response process.

Step 4: Recovery Actions

After the attack is controlled, normal network operations are restored and system performance is verified.

Step 5: Post-Incident Review

The incident is reviewed to analyze the attack method, evaluate the response effectiveness, and update the playbook if required.

Expected Results

A well-structured DDoS incident response playbook is created, enabling quicker and more effective response to future attacks.

Conclusion

Using an incident response playbook helps organizations respond to DDoS attacks efficiently by reducing response time and minimizing human errors during critical situations.